

Research Proposal Summary

Title

Evolutionary Functions of Infant-Directed Speech: Impact on Attention, Auditory Preferences, and Early Language and Musical Development [Funciones evolutivas del habla dirigida a bebés: Impacto en la atención, las preferencias auditivas y el desarrollo lingüístico y musical temprano]

Total budget

COP 639,928,324 (≈ USD 161,000, for the 30-month project)

Executive Summary

Infant-Directed Speech (IDS) is a universal form of vocal communication that plays a crucial role in the development of prelinguistic infants. It is essential for infant wellbeing and development: it facilitates mother-infant attachment by modulating oxytocin and other neuropeptides¹⁻³, supports language acquisition⁴⁻⁹, regulates temperament^{10,11}, and coordinates communicative interactions¹²⁻¹⁴. IDS also provides insights into the origins of language and the evolution of music^{15,16}. Acoustic features of IDS, such as exaggerated intonation and high pitch variability, capture infant attention^{17,18}. Moreover, infants' attention to a speaker's mouth predicts later vocabulary growth¹⁹⁻²².

This project will investigate the impact of acoustic features of IDS on infants' attention to vocal stimuli, and their relationship to language and musical development. We will conduct two complementary studies:

1. Using eye-tracking, we will examine how experimental acoustic manipulations influence the attention of 3- to 9-month-old infants toward the speaker.
2. We will analyze the acoustic properties of IDS produced by mothers of children aged 16 to 23 months to assess their predictive value for later language and musical development.

Problem Statement

Humans are a unique species: among many traits, we speak and sing. These profoundly human capacities are key to the emergence and transmission of cultural phenomena and depend on extensive, complex social learning. From birth, infants rely on communication with their caregivers for survival. Although they cannot yet speak or understand language, they need to communicate to strengthen emotional bonds, avoid neglect, express and interpret emotions, and begin the process of language acquisition.

A universal process underlies these early interactions: Infant-Directed Speech (IDS). Maternal IDS is not a curiosity or cultural peculiarity; it is one of the earliest and most powerful forms of social interaction, with a decisive influence on human development¹⁵.

Vocal communication is widespread among tetrapods, but in humans, nonverbal vocal parameters (such as fundamental frequency, f_0 , and vocal tract resonances, formant dispersion Df) communicate biologically and socially relevant information. These paralinguistic cues allow recognition of individuals and the assessment of traits such as sex²³, body size²⁴⁻²⁸, attractiveness^{27,29-34}, sexual maturity³⁵, or social power³⁶. Humans also actively modulate their voices depending on social context, expressing emotions and motivations, and influencing how they are perceived by others³⁷. Voice modulation thus carries adaptive social implications^{38,39}.

IDS is one of the most prominent forms of voice modulation in humans. It is characterized by higher pitch, greater pitch variability, exaggerated intonation, slower tempo, clearer articulation, and positive emotional tone. These features attract infants' attention, optimize the transmission of social and linguistic signals, and are observed across languages and cultures⁴⁰⁻⁴⁵. IDS fosters attachment, regulates emotion, supports joint attention, and predicts later language comprehension^{4,5}. Its emotional qualities, especially when attuned to the infant's state, enhance mother-infant synchrony. High-quality IDS is linked to better maternal mood and lower parental stress, while depression can reduce both quantity and expressivity of IDS, affecting attachment and child development^{42,46}.

IDS not only promotes language acquisition^{6,8,47} by highlighting prosodic cues, segmenting speech^{45,48,49}, and facilitating vocabulary learning, but also plays a role in emotional regulation. For example, in stressful contexts, infants regulate negative emotions better when exposed to IDS with typical prosody⁵⁰. Singing directed to infants may be even more effective than speech in reducing distress, suggesting that musical elements of IDS contribute to emotional regulation^{51,52}.

Despite its significance, experimental studies of how acoustic properties of IDS influence infant attention are scarce, and no studies have tested whether maternal IDS predicts later linguistic (and possibly musical) skills. This project seeks to address these gaps, advancing knowledge on the developmental and evolutionary functions of IDS.

Justification

1. **Research Gaps:** IDS has well-documented roles in attachment, attention, emotion regulation, and language development. However, little is known about the specific effects of its acoustic properties on infant attention or its predictive power for later developmental outcomes.
2. **Social and Educational Relevance:** By addressing a universal phenomenon—how caregivers speak to infants—this project informs early childhood education, parenting

strategies, and public health interventions, particularly in contexts of maternal depression or communication challenges.

3. **Scientific and Technical Relevance:** Positioned at the intersection of developmental science, evolutionary biology, and language neuroscience, the project combines acoustic analyses, eye-tracking, and longitudinal designs. It adheres to open science principles, including preregistration, simulation-based power analysis, open-source software, and data sharing. Including participants from the Global South helps counterbalance representational biases in behavioral sciences.

Conceptual Framework

1. **Vocal Modulation:** Humans, like many species, modulate vocalizations according to context. Our exceptional vocal control supports speech, singing, and expressive nonverbal vocalizations (e.g., laughter, roars). Such modulations convey emotion and can strategically influence social perception.
2. **Infant-Directed Speech:** IDS is a universal, intentional form of vocal modulation distinguished by exaggerated acoustic features. It is adaptive for infant attention, emotional regulation, and language learning. Cross-cultural evidence shows IDS and infant-directed singing are identifiable even across societies and capture infants' attention more effectively than adult-directed speech.
3. **IDS, Language, and Music:** IDS provides insights into the evolutionary roots of both language and music. Some models propose a common proto-musical language, with IDS situated at the intersection of emotional and referential communication. Lullabies and playful songs, nearly universal, calm infants and support affect regulation.
4. **Evolutionary Model:** We propose IDS evolved under selective pressures linked to prolonged infant dependency, facilitating attachment and survival. Its link to oxytocin release and emotional bonding supports this idea. IDS may also signal parental quality, overlapping with sexual selection processes shaping musicality and vocal expressiveness.
5. **Experimental Opportunity:** We predict that infants will prefer IDS with greater pitch variability—a key acoustic feature—but this remains untested. Modern methods (precise acoustic manipulation, eye-tracking, facial and physiological analysis) allow unprecedented testing of these hypotheses.

General Objective

To analyze how maternal IDS acoustic features influence early social attention in prelinguistic infants, and whether these features predict later language and musical development.

Specific Objectives

1. Characterize the acoustic properties of maternal IDS toward infants aged 3–9 months.
2. Experimentally test the effect of IDS features on infant social attention using controlled auditory stimuli and eye-tracking.
3. Analyze infants' emotional and physiological responses to IDS manipulations using AI-based facial and physiological measures.
4. Examine whether maternal IDS properties during the first year predict children's language and musical abilities at 16–23 months.
5. Explore associations between language and musical development and early exposure to IDS.
6. Further develop and validate an open-source webcam-based eye-tracker for remote infant studies.
7. Disseminate knowledge on IDS through accessible digital communication strategies targeting caregivers, educators, and health professionals.
8. Generate applied materials transferable to maternal-child health and early education contexts.

Methodology

Study 1: Acoustic Manipulation of IDS and Infant Attention

Participants: Infants aged 3–9 months capable of holding their heads upright.

Stimuli: 5-second IDS samples recorded from mothers, manipulated along three independent acoustic parameters:

- **Mean pitch (f_0 mean):** shifted ± 20 Hz.
- **Pitch variability (f_0 SD):** shifted ± 10 Hz.
- **Vocal tract length (formant dispersion, Df):** shifted $\pm 10\%$.

These manipulations yield 8 experimental conditions (2x2x2 design).

Procedure:

- Infants will be seated on caregivers' laps in front of a screen.
- Stimuli will be presented in randomized blocks, separated by short pauses to maintain attention.

- Eye-tracking will record gaze direction, fixation durations, and saccade latencies (>100 Hz sampling, <0.5° accuracy).
- Synchronized video will capture infants' facial expressions, body movements, and skin color changes.
- AI tools will analyze videos for emotional valence/activation and estimate heart rate via remote photoplethysmography.

Sample size and power:

- Monte Carlo simulations (10,000 infants) modeled effects of f_0 SD, f_0 mean, and Df.
- Results: $N \approx 160$ infants provide >80% power for main effects, with ~98% detection probability for f_0 SD.

Duration: Each session ~30 minutes. Participation ends immediately if infants show discomfort or fall asleep. Transportation stipends will be provided to families.

Study 2: Maternal IDS Predicting Later Language and Musical Development

Participants: ≥ 480 mothers of children aged 16–23 months.

Data Collection:

1. Upload of home videos showing maternal IDS during infants' first year.
2. Completion of online questionnaires (Spanish versions):
 - **CDI-2 (Words and Sentences)** for language.
 - **Music@Home-Infant** for musical environment and behaviors.
3. Demographic data: maternal age, education, SES, family structure, and infant age.

Analyses:

- Acoustic features (f_0 mean, f_0 SD, Df) extracted from IDS.
- Outcomes: CDI and Music@Home scores (adjusted for covariates via regression residualization).
- Power: With $N=480$, $\geq 90\%$ power to detect correlations of $\rho=0.30$ or conclude statistical equivalence within ± 0.15 .

Interpretation:

- Significant positive associations $\rightarrow$ evidence that IDS predicts developmental outcomes.
- Equivalence test results $\rightarrow$ evidence that effects are trivial or absent.

Ethical Considerations

Both studies involve non-invasive methods and are classified as minimal risk. Infant participation will be based on parental consent, and sessions will be terminated if any signs of discomfort appear. Data will be anonymized and stored securely. A prior ethics approval (CIE 2023-059) covers similar protocols, and this project will seek renewed approval. Special care will be taken to ensure infant welfare and voluntary participation.

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